2026

$2 + 0 + 2 + 6 = 10 \to 1$

so 2026 leaves remainder 1 when divided by 9

Numbers That Talk

Divisibility rules and why they work

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Is 2026 divisible by 9?

Magic trick 1: the vanishing digit

1. Think of a 4-digit number (not all digits the same).
2. Shuffle its digits into a new number.
3. Subtract: bigger minus smaller.
4. Circle one non-zero digit of the result. Hide it.
5. Read me the other digits, in any order.

example

5274 → **7425**

7425 − 5274 = **2 1 5 1**

They read: 2, 1, 1 → sum 4

4 needs 5 to reach 9. The hidden digit is 5.

If the sum is already a multiple of 9, the answer is 9.

Write along in box 1 of your worksheet

A clock with only 9 marks



walk 10 steps = walk 1 step

$$2026 = 2\cdot1000 + 0\cdot100 + 2\cdot10 + 6$$

$$= 2\cdot(999 + 1) + 0\cdot(99 + 1) + 2\cdot(9 + 1) + 6$$

$$= (2\cdot999 + 0\cdot99 + 2\cdot9) + (2 + 0 + 2 + 6)$$

a multiple of 9 + the digit sum

$$\text{so } 2026 \equiv 2 + 0 + 2 + 6 = 10 \equiv 1 \pmod 9$$

Same digits $\Longrightarrow$ same digit sum $\Longrightarrow$ same remainder.

Subtract $\Longrightarrow$ remainder 0, so the answer's digits add up to a multiple of 9.

What about 37?

$$1000 = 999 + 1 \implies 1000 \equiv 1 \pmod{999}, \quad 999 = 27 \times 37$$

So cut a number into three-digit chunks from the right and add them. That tests 27 and 37.

45658 → **45** + **658** = 703 = 19 × 37 ✓

37 divides 45658, no long division. (It is 37 × 1234. We checked.)

Always chunk from the right: 45 | 658, not 456 | 58

Family one: $10^k \equiv 1$

$10 \equiv 1 \pmod 9$ → digit sum

$100 \equiv 1 \pmod{99}$ → 2-digit chunks

$1000 \equiv 1 \pmod{999 = 27 \times 37}$ → 3-digit chunks

7 ? 13 ?

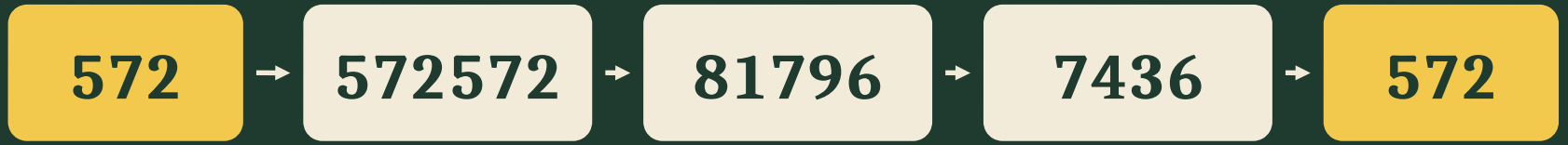
999 = 27 × 37. No 7, no 13 anywhere.

Does 7 even have a rule?

What if $10^k \equiv -1$ instead?

over to Yueran

Magic trick 2: 1001 brings it back



Every division comes out exact. The last one gives back the number you started with. Try your own number.

box 2 of your worksheet

_____ → _____ → ÷ 7 → ÷ 11 → ÷ 13 → _____

Why trick 2 works

$$abcabc = abc \cdot 1000 + abc = abc \cdot 1001 = abc \cdot 7 \cdot 11 \cdot 13$$

And $1000 = 1001 - 1$, so $1000 \equiv -1 \pmod{1001}$.

Cut into three-digit chunks from the right and alternate + − + −. You get the remainder mod 7, 11 and 13 at once.

2026 → **026** − **2** = 24

$24 \equiv 3 \pmod{7}$

$24 \equiv 11 \pmod{13}$

Both are the true remainders of 2026.

572572 → 572 − 572 = 0, so 7, 11 and 13 all divide it. That's the trick again.

Does every prime have a rule?

find k with $10^k \equiv +1$ or $-1 \pmod d$

+1: add k-digit chunks. 9 (k = 1), 37 (k = 3)

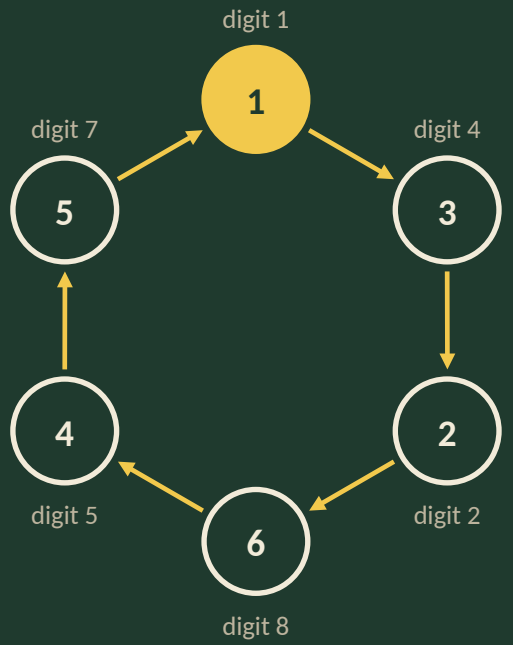
−1: alternate them. 7 and 13 (k = 3)

Fermat: $10^{p-1} \equiv 1 \pmod p$ for every prime $p \neq 2, 5$

So a rule always exists.

Shortest additive chunk = $\text{ord}_p(10)$ = the period of 1/p.

$1/7 = 0.142857\ 142857\ \ldots$ $\quad \text{ord}_7(10) = 6$



remainders of 1 ÷ 7 = powers of 10 mod 7

142857 × 2 = 285714, × 3 = 428571

If you know what 10 is mod d, you can build the rule for d yourself.

Fermat says the rule exists. The order of 10 says how short it can be.

The proofs are on the back of your worksheet.

There is more (a quick rule for 7, why ISBN numbers use 11). Ask us after class!

Thank you. Now go try the 1001 trick on a friend.